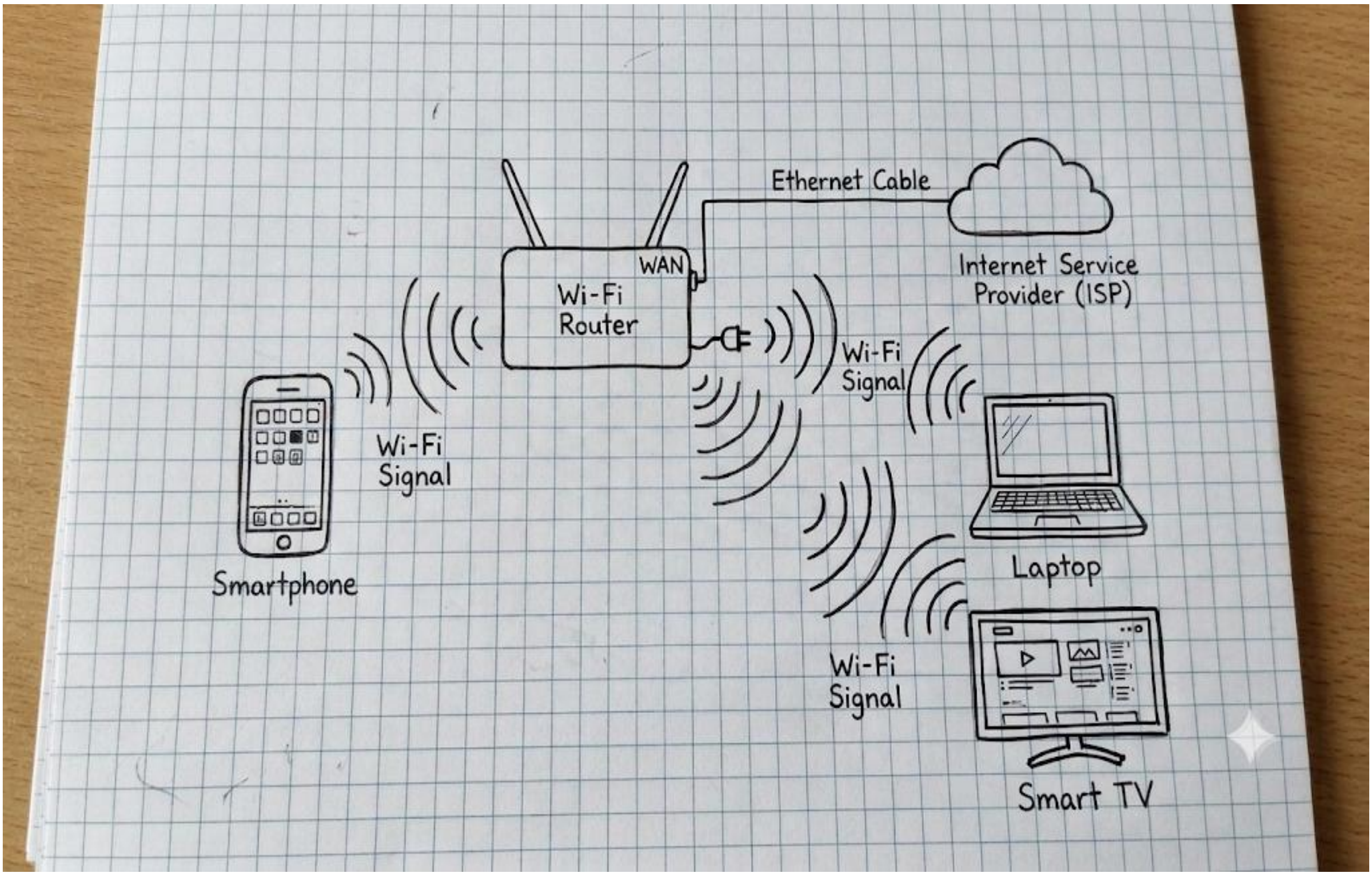
Network Devices

1. Connection Of Network :



2. Network Devices Comparison Table :

Network Device	Primary Role & Layer	Real-World Use Case	Daily App / Service Example
Router	Connects different networks together and routes data using IP addresses (Layer 3).	Links your private home network to your ISP's network to share a single internet connection.	**YouTube Streaming:** Routes incoming video packets from YouTube servers to your home network.

Network Device	Primary Role & Layer	Real-World Use Case	Daily App / Service Example
Switch	Connects multiple devices within the same local network using MAC addresses (Layer 2).	Connects multiple PCs and consoles together via cables in a room without slowing down traffic.	**LAN Gaming (Forza / GTA):** Allows multiple PCs in a gaming cafe to share files and play local multiplayer smoothly.
Hub	Legacy device that broadcasts incoming data frames to *all* connected ports (Layer 1).	Simple multi-device connectivity for basic low-traffic hardware, though rarely used today due to collisions.	**Basic Smart Home Sensors:** Sending tiny status updates to a legacy hub line.
Access Point (AP)	Extends local network access wirelessly via radio waves (Layer 2).	Provides high-speed wireless connectivity across large areas like college campuses or multi-story homes.	**Zomato / Instagram:** Keeps your smartphone connected to local Wi-Fi while moving around different rooms.

3. Ordering Food on Zomato – Path of Data :

When you open **Zomato** on your smartphone and tap *"Place Order"*, your data travels through these network devices:

1. **Smartphone to Access Point (Built into Router):** Your phone converts the Zomato checkout request into radio waves and sends it wirelessly to the Access Point inside your home Wi-Fi router.
2. **Access Point to Router Switch Module:** The Access Point converts the wireless radio signals back into digital electrical signals and passes them to the internal switch portion of the router.

3. **Router to Internet Service Provider (ISP):** The Router assigns your request a public IP address (using NAT) and routes the data packets through the WAN port onto the internet backbone toward Zomato's servers.
4. **Return Path:** When Zomato sends back an order confirmation, the **Router** receives it from the internet, the internal **Switch** forwards it to the **Access Point**, and the AP broadcasts it back to your **Smartphone**.

4. Connecting a Far away Gaming PC to a Wi-Fi Router :

Wi-Fi Extender / Range Extender (with Ethernet Port)

- **How it works:** A Wi-Fi extender receives the wireless signal from your main router, boosts it, and converts it into a wired connection through its onboard LAN port.
- **Setup:** Plug the extender into an electrical outlet near your gaming PC where Wi-Fi signal is decent. Connect an Ethernet cable from the extender's LAN port directly into your PC's Ethernet port.
- **Benefit:** Functions as an external wireless adapter for your PC, giving it instant internet access.